AWS SAA-C03 Study Note: Storage Extras

1. AWS Snow Family Overview

The Snow Family is used for **offline** data migration and edge computing. It is the answer when network capacity is insufficient for the volume of data (e.g., "it would take years to transfer over the internet").

Device Comparison

Device	Capacity (Usable)	Key Features	Exam Use Case
Snowcone	8 TB (HDD) or 14 TB (SSD)	Portable, rugged, battery option. Fits in a backpack.	Small data transfers (e.g., 5TB) from rough environments (deserts, oil rigs) or individual IoT collection points.
Snowball Edge Storage Optimized	80 TB	High storage capacity. 40 vCPUs.	Large-scale data migration (petabytes). Replacing physical tape libraries.
Snowball Edge Compute Optimized	42 TB	More compute power (52 vCPUs, GPU options).	Edge processing (ML inference, video analysis) *before* sending data to AWS.
Snowmobile	100 PB	Shipping container truck.	Exabyte-scale migrations (e.g., entire datacenter shutdown).

Key Exam Rule: If the transfer takes **more than a week** over the network, consider Snowball.

2. Architecture: Snowball into Glacier

A common exam scenario asks for the most cost-effective way to migrate petabytes of archives from on-premises directly into cold storage (Glacier).

The "Lifecycle" Pattern (Classic)

1. **On-Premises:** Load data onto Snowball Edge.
2. **Transport:** Ship device to AWS.
3. **Ingest:** AWS imports data into an **S3 Standard** bucket.
4. **Transition:** Use an **S3 Lifecycle Policy** to transition objects to S3 Glacier Flexible Retrieval or Deep Archive immediately (0 days).
    - *Drawback:* You pay a transition fee per object.

The "Direct Import" Pattern (Optimized)

Snowball Edge now supports importing data **directly** into S3 Glacier storage classes.

- **How:** When creating the job in the console, select the destination storage class as **Glacier Flexible Retrieval** or **Glacier Deep Archive**.
- **Benefit:** Avoids the S3 Lifecycle transition costs. This is often the "most cost-effective" answer for archival migration.

3. Amazon FSx

FSx provides fully managed third-party file systems. You must match the "Operating System" or "Workload" to the FSx type.

Service	Protocol	OS / Compatibility	Primary Exam Use Case
FSx for Windows File Server	SMB	**Windows**	"Lift and shift" Windows apps. Shared storage for Windows servers. **Active Directory** integration. Distributed File System (DFS).
FSx for Lustre	POSIX	**Linux**	**HPC** (High-Performance Computing), Machine Learning, Video Processing. "Fast parallel file system."
FSx for NetApp ONTAP	NFS, SMB, iSCSI	Linux & Windows	Migrating on-prem **NetApp** workloads. Supports de-duplication, compression, and snapshotting.
FSx for OpenZFS	NFS	Linux	Migrating ZFS workloads. High performance (1 million IOPS).

FSx for Lustre Deployment Types (Crucial)

The exam will often ask you to choose between these two modes based on data durability needs and job length.

Feature	Scratch File System	Persistent File System
Use Case	Temporary storage, short-term processing (weeks/months).	Long-term storage, sensitive to availability.
Durability	**NOT replicated.** If a server fails, data is lost.	**Replicated** within the Availability Zone (AZ). Self-healing if a server fails.
Performance	Higher burst throughput (up to 6x baseline).	Consistent baseline performance.
Cost	Lower cost.	Higher cost.

Feature	Scratch File System	Persistent File System
Exam Trigger	"Temporary processing," "Cost optimization," "Data can be reproduced from S3."	"High availability," "Long-term storage," "Data cannot be easily reproduced."

FSx for Lustre + S3 Integration

- You can link an FSx for Lustre file system to an S3 bucket.
- The S3 objects appear as files in the file system.
- **Workflow:**
  1. Spin up a **Scratch** FSx for Lustre file system linked to S3.
  2. Process data at high speed.
  3. Write results back to S3.
  4. **Delete** the FSx file system to save money.

4. AWS Storage Gateway Overview

Hybrid storage service that enables on-premises applications to use AWS cloud storage. It requires a virtual or hardware appliance installed on-premises.

1. S3 File Gateway

- **Interface:** NFS or SMB.
- **Backend:** Stores files as **Objects** in an S3 bucket.
- **Use Case:** User home directories, data lakes, content repositories that need to be accessible as files on-prem but stored cheaply in S3.
- **Caching:** Keeps "recently accessed" data locally for low latency.

2. FSx File Gateway

- **Interface:** SMB.
- **Backend:** Amazon **FSx for Windows File Server**.
- **Use Case:** Low-latency on-prem access to Windows file shares in the cloud. Extension of FSx.

3. Volume Gateway (Block Storage / iSCSI)

Presents disk volumes (iSCSI) to on-prem servers.

- **Stored Volumes:** All data is stored **locally** (on-prem) for maximum speed. Asynchronous backups are sent to AWS (EBS Snapshots).
  - *Use Case:* fast local access + offsite backup.
- **Cached Volumes:** Primary data is in **AWS (S3)**. Only frequently accessed data is cached locally.

- *Use Case:* expanding on-prem storage cheaply without buying new hardware.

4. Tape Gateway

- **Interface:** iSCSI VTL (Virtual Tape Library).
- **Backend:** S3 (Virtual Tapes) -> Glacier / Deep Archive.
- **Use Case:** Killing physical tape infrastructure. Backup software (e.g., Veeam, Veritas) thinks it's writing to physical tapes, but it's going to AWS.

5. AWS Transfer Family

A serverless service to manage file transfers *into* and *out of* AWS storage without managing infrastructure.

- **Protocols Supported:**
  - **SFTP** (Secure File Transfer Protocol) - *Most tested.*
  - **FTPS** (File Transfer Protocol over SSL)
  - **FTP** (Unencrypted - rarely the answer unless "legacy support" required)
- **Destinations:** Amazon S3 or Amazon EFS.
- **Exam Scenario:** "A 3rd party partner needs to upload weekly reports via SFTP. We do not want to manage EC2 instances for this." -> **AWS Transfer Family**.

6. AWS DataSync - Overview

An online data transfer service designed to automate moving data between on-premises storage and AWS.

- **Mechanism:** Requires a **DataSync Agent** (VM) installed on-premises.
- **Protocol:** Uses a proprietary, accelerated network protocol (not standard traffic) over Internet or Direct Connect.
- **Speed:** Much faster than standard command-line tools (like AWS CLI `cp`).
- **Destinations:** S3, EFS, FSx for Windows, FSx for Lustre.
- **Scheduling:** Supports bandwidth throttling and scheduling (e.g., "transfer only at night").
- **Exam Scenario:** "Migrating a large on-prem NFS server to EFS over the weekend" or "Ongoing replication of on-prem data to AWS."

7. All AWS Storage Options Compared (Cheat Sheet)

Scenario	Best Service	Why?
Object Storage (Images, Videos, Backups)	**S3**	Limitless, serverless, web-accessible.
Block Storage (OS Boot Volume, Database)	**EBS**	Low latency, attached to one EC2 instance.

Scenario	Best Service	Why?
Shared File Storage (Linux EC2 instances)	**EFS**	POSIX compliant, scales automatically, multi-AZ.
Shared File Storage (Windows EC2 instances)	**FSx for Windows**	Native SMB, AD integration.
HPC / Parallel File System	**FSx for Lustre**	High throughput, links to S3.
Local Cache + Cloud Backed Files	**S3 File Gateway**	NFS/SMB interface on-prem, data lives in S3.
Local Cache + Cloud Backed Block	**Volume Gateway**	iSCSI interface, extends on-prem disks to cloud.
Replace Physical Tapes	**Tape Gateway**	VTL interface, cheap archive in Glacier.
Offline Migration (Petabytes)	**Snowball Edge**	No internet bandwidth used.
Online Migration (NFS/SMB to AWS)	**DataSync**	Automated, accelerated, agent-based.
Partner SFTP Uploads	**Transfer Family**	Fully managed SFTP endpoint.

8. Simplified Decision Tree (Memory Aids)

Use these short "If-Then" rules to answer questions quickly.

Rule 1: Moving Data to AWS

- **No Internet** (or "weeks to transfer")? → **Snowball**
- **Has Internet** (Fast/Direct Connect)? → **DataSync**
- **SFTP / FTP Protocol?** → **Transfer Family**

Rule 2: Hybrid Access (Storage Gateway)

- **Need to see FILES?** (NFS/SMB) → **S3 File Gateway**
- **Need a DISK?** (iSCSI) → **Volume Gateway**
- **Need to Replace TAPES?** → **Tape Gateway**

Rule 3: FSx Keywords

- **Windows / Active Directory / SMB** → **FSx for Windows**
- **Linux / HPC / "Scratch"** → **FSx for Lustre**
- **NetApp** → **FSx for ONTAP**

Rule 4: Lustre Types

- **Fast & Temporary** → **Scratch** (Not replicated, cheaper)
- **Long-term & Safe** → **Persistent** (Replicated, higher availability)